

Q-2:

Implement a C++ program using object-oriented principles to count the occurrences of a specific element in an array. Design a template function that works for arrays of int, float, double, and char types. Your program should include the following:

1. Define a template function **CountOccurrences** that takes an array and the size of the array as arguments, along with a target element to count.
2. Implement the **CountOccurrences** function using a loop to iterate through the array elements, count the occurrences of the target element and return the count.
3. Write a **main** function that demonstrates the usage of the **CountOccurrences** function by creating arrays of different types (**int**, **float**, **double**, and **char**) with varying sizes. Specify a target element for each array and call the **CountOccurrences** function to count the occurrences.
4. Display the number of occurrences of the target element for each array type.
5. Test your program with different arrays and target elements to verify its correctness and robustness.

Solution:

```
#include <iostream>
using namespace std;

template<typename T, size_t N>

size_t CountOccurrences( const T (&arr)[N], const T& target )
{
    size_t count = 0;
    for (size_t i = 0; i < N; ++i) {
        if (arr[i] == target) {
            count++;
        }
    }
    return count;
}

int main() {
    int intArr[] = {1, 2, 3, 4, 5, 2};
    float floatArr[] = {1.1f, 2.2f, 3.3f, 2.2f, 4.4f};
    double doubleArr[] = {1.1, 2.2, 3.3, 4.4, 5.5, 2.2};
    char charArr[] = "Hello, World!";

    int intTarget = 2;
    float floatTarget = 2.2f;
    double doubleTarget = 2.2;
    char charTarget = 'o';

    size_t intCount = CountOccurrences(intArr, intTarget);
    size_t floatCount = CountOccurrences(floatArr, floatTarget);
    size_t doubleCount = CountOccurrences(doubleArr, doubleTarget);
    size_t charCount = CountOccurrences(charArr, charTarget);

    cout << "Occurrences of " << intTarget << " in int array: " <<
intCount << endl;
    cout << "Occurrences of " << floatTarget << " in float array: " <<
floatCount << endl;
    cout << "Occurrences of " << doubleTarget << " in double array: " <<
doubleCount << endl;
    cout << "Occurrences of '" << charTarget << "' in char array: " <<
charCount << endl;

    return 0;
}
```